



Hybrid Pipeline for Moroccan Dialect Classification: Oujdi (Eastern Darija) vs Standard Darija using TF-IDF, BiLSTM, and DistilBERT with LoRA

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Abstract:

Moroccan Darija is a rich and highly variable spoken language whose regional sub-dialects remain poorly represented in computational linguistics. In this work, we focus on the Oujdi dialect, spoken in the Eastern region of Morocco around the city of Oujda, and we design a complete pipeline to automatically distinguish Oujdi text from standard Moroccan Darija.

We construct a balanced bi-script dataset of 100,000 sentences (50,000 Oujdi, 50,000 Moroccan), built from the DODa corpus for standard Darija and a 220,000-line Oujdi corpus, with samples written in both Arabic script and Latin (Arabizi) script. We train and compare three families of classifiers under identical splits: a TF-IDF character-level Logistic Regression baseline, a Bidirectional LSTM with learned embeddings, and a multilingual DistilBERT encoder fine-tuned with Low-Rank Adaptation (LoRA).

Our experiments show that the DistilBERT + LoRA configuration reaches the best test accuracy of 97.59%, while updating less than 1% of the original parameters. The model maintains strong performance across both Arabic and Latin scripts, demonstrating that parameter-efficient multilingual transformers are well suited to fine-grained regional dialect identification. We further package the best model behind a Flask API and a React user interface to make the system practically usable.

Keywords: Moroccan Darija, Oujdi dialect, Arabic dialect identification, code-switching, Arabizi, character n-grams, BiLSTM, multilingual DistilBERT, LoRA, parameter-efficient fine-tuning.

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The code associated with this study is available on GitHub:

[github.com/chamail/
classification-dialect-oujdi](https://github.com/chamail/classification-dialect-oujdi)

1 Introduction

Arabic is not a single language but a continuum of dialects that differ substantially across regions. Within Morocco alone, the spoken language — locally referred to as *Darija* — varies noticeably between cities, and the Eastern region around Oujda has developed a distinctive sub-dialect known as *Oujdi*, influenced both by the rest of Moroccan Darija and by neighbouring Algerian varieties. While Moroccan Darija has begun to receive attention in natural language processing, particularly through resources such as the DODa corpus, regional sub-dialects like Oujdi remain largely absent from existing datasets and benchmarks.

Automatically identifying the dialect of a short text is a foundational task for any downstream pipeline that needs to handle Moroccan content reliably: machine translation, sentiment analysis, content moderation, voice assistants, and conversational agents all benefit from knowing which variety they are processing. This problem becomes more difficult in the Moroccan context because users freely switch between two writing systems — Arabic script and Latin script with digits (commonly called *Arabizi*) — often within the same message, and because the vocabulary overlap between Oujdi and standard Darija is high.

Recent work in Arabic dialect identification has shown that transformer-based encoders pretrained on multilingual or Arabic corpora can capture subtle lexical and morphological cues that distinguish related varieties [2]. At the same time, parameter-efficient fine-tuning techniques such as Low-Rank Adaptation (LoRA) [5] have made it realistic to adapt these encoders on modest hardware, which is important for student research and for regional dialect projects that cannot rely on large compute budgets.

In this work, we focus on the binary classification problem of distinguishing Oujdi from standard Moroccan Darija in short user-style sentences. Our objective is twofold: (i) to build a clean and balanced bi-script dataset for this specific pair, since none was publicly available, and (ii) to evaluate how three different families of models behave on the same data: a strong classical baseline based on character n-grams and Logistic Regression, a Bidirectional LSTM trained from scratch, and a multilingual DistilBERT encoder fine-tuned with LoRA.

A key concern throughout this study is that the writing system itself should not dictate the prediction. Because both classes contain a large proportion of Arabic-script and Latin-script samples in roughly equal numbers, the model must learn lexical and sub-word cues, not surface alphabet statistics. We discuss how character-level and sub-word tokenization choices interact with this constraint.

The remainder of this paper is organized as follows. Section 2 reviews related work on Arabic dialect identi-

fication and parameter-efficient fine-tuning. Section 3 describes the data sources and the construction of our 100,000-sentence corpus. Section 4 details the three models, their hyperparameters, and the LoRA configuration. Section 5 reports the comparative results on the test set. Section 6 concludes the paper and outlines future work.

2 Related Work

This section reviews previous studies relevant to our problem, organized around three themes: Arabic dialect identification, resources for Moroccan Darija, and parameter-efficient fine-tuning.

Arabic dialect identification. Zaidan and Callison-Burch introduced one of the earliest large-scale studies on Arabic dialect identification, distinguishing Modern Standard Arabic from four regional varieties (Egyptian, Levantine, Gulf, Maghrebi) on a newspaper-comment corpus. Their work established that simple word and character n-gram features already carry strong dialectal signal, and it motivated the construction of dedicated dialectal corpora rather than relying on MSA-only resources [1]. Many subsequent benchmarks build on this insight at the country and city level.

Deep learning for fine-grained dialects. Abdul-Mageed et al. extended dialect identification to a fine-grained, city-level setting with the NADI shared tasks, showing that pretrained Arabic transformer encoders outperform classical n-gram models on short, noisy social media text, especially when target classes share a substantial portion of their vocabulary [2]. Their findings directly motivate our use of a multilingual transformer for the Oujdi versus Darija pair, where lexical overlap is high.

Resources for Moroccan Darija. Outchakoucht and Es-Samaali released DODa (Darija Open Dataset), a multi-purpose corpus containing parallel sentences and lexicon entries for Moroccan Darija in both Arabic and Latin scripts. DODa is, to our knowledge, the largest publicly available Moroccan Darija resource and is the basis for the Moroccan side of our dataset [3]. However, DODa treats Darija as a single class and does not annotate regional sub-dialects, which is precisely the gap we address by pairing it with an Oujdi-specific corpus.

Parameter-efficient fine-tuning. Hu et al. introduced Low-Rank Adaptation (LoRA), a method that injects small trainable low-rank matrices into the attention projections of a frozen pretrained transformer. LoRA dramatically reduces the number of trainable parameters — often to less than 1% of the original model — while matching or approaching full fine-tuning performance [5]. This makes LoRA especially attractive for student projects and regional NLP tasks where compute and memory are limited, and it is the strategy we adopt for our DistilBERT-based configuration.

Overall, these works confirm two trends that guide

our design: (i) for closely related Arabic varieties, transformer-based encoders are the strongest backbone, and (ii) parameter-efficient adaptation is sufficient to specialize a generic multilingual model on a focused dialect-pair task without retraining from scratch.

3 Datasets

3.1 Data Collection

Our dataset is constructed from two complementary sources, one per target class.

The **Moroccan Darija** class is drawn from the DODa corpus [3], which provides parallel sentences in Moroccan Darija written in both Arabic script (`darija_ar`) and in Latin script (`darija`). We extract both columns so that the resulting class is balanced across the two writing systems and reflects the way Moroccans actually write online.

The **Oujdi** class is drawn from an Oujdi-specific corpus containing approximately 220,000 sentences in JSON Lines format, with each entry providing the same content in Arabic script (`output`) and in Latin transliteration (`wjdi_lat`). As for Darija, we pool both fields so that the Oujdi class is also represented in both scripts.

Both classes are therefore intentionally bi-script. This is essential, because if one class were predominantly in Arabic script and the other in Latin script, any classifier could trivially solve the task by detecting the alphabet rather than the dialect.

3.2 Data Preparation

Each raw sentence goes through three preprocessing steps:

1. **Normalization.** Leading and trailing whitespace is stripped, and internal whitespace is collapsed into single spaces.
2. **Validity filtering.** A sentence is kept only if its length is between 3 and 150 characters and if it contains at least one Arabic character (Unicode range U+0600-U+06FF) or one Latin character. Empty strings and lines containing only punctuation or digits are discarded.
3. **Deduplication.** Exact textual duplicates are removed within each class before sampling, to prevent train/test leakage.

After filtering, we sample exactly 50,000 sentences per class with a fixed random seed (`random_state=42`), so the final dataset is perfectly balanced. We then add an auxiliary `script` column that labels each sentence as `arabic`, `latin`, or `mixed`, depending on which character ranges occur. Table 1 summarizes the resulting dataset.

Table 1: Summary of the constructed dataset.

Field	Count
Oujdi sentences	50,000
Moroccan Darija sentences	50,000
Arabic script	49,839
Latin script (Arabizi)	50,034
Mixed script	127
Total	100,000

Figure 1 confirms the perfectly balanced binary distribution between the two classes, and Figure 2 shows the distribution of sentence lengths per class. Sentence lengths are concentrated in the short range (typical of social-media style messages), with no class showing an abnormally long tail, which justifies the choice of a moderate maximum sequence length (128 sub-word tokens) for the transformer model.

Dataset Class Distribution

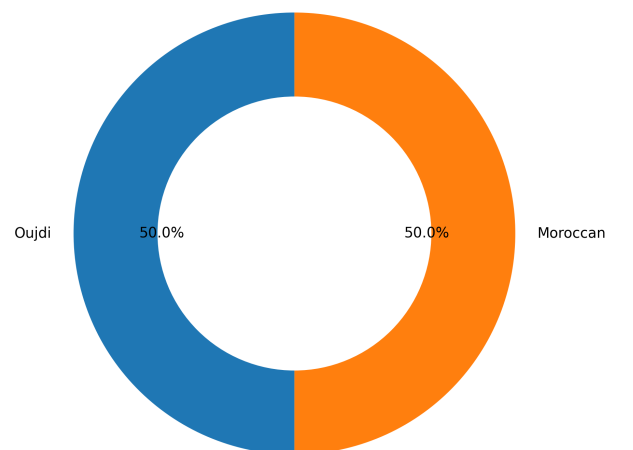


Figure 4 : Balanced dataset distribution with 50,000 samples per class.

Figure 1: Class distribution of the final corpus: 50,000 Oujdi and 50,000 Moroccan Darija sentences, perfectly balanced.

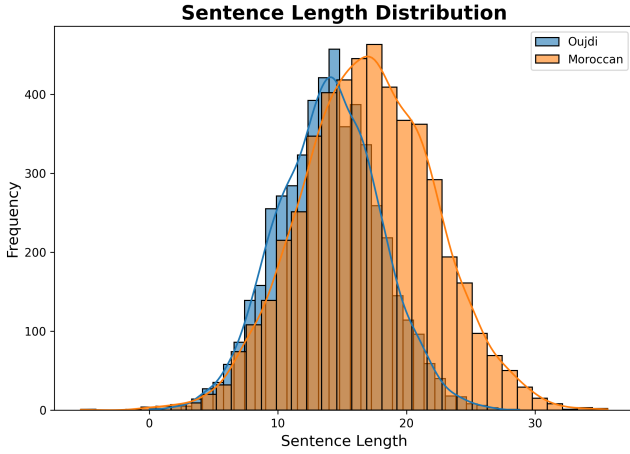


Figure 15 : Sentence length distribution per dialect.

Figure 2: Sentence length distribution per dialect. Both classes follow similar short-text profiles, characteristic of user-generated content.

Finally, we split the data into training, validation, and test sets following the proportions used by the strongest model (Section 4). For the baseline and BiLSTM we use a standard 80/20 train/test split with stratification on the class label, while for the DistilBERT + LoRA configuration we further split the 20% holdout into validation and test (10% / 10%) to allow checkpoint selection. Table 2 summarizes the splits for the transformer configuration.

Table 2: Data split used for the DistilBERT + LoRA configuration.

Split	Oujdi	Moroccan
Train	40,000	40,000
Validation	5,000	5,000
Test	5,000	5,000
Total	50,000	50,000

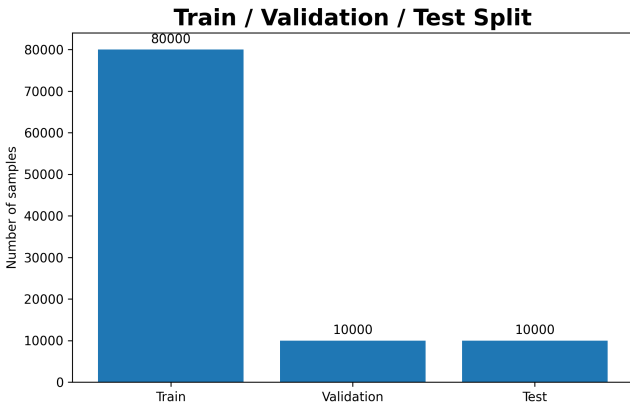


Figure 5 : Dataset split used for model training and evaluation.

Figure 3: Visualization of the train / validation / test split used to train and evaluate the transformer model.

4 Methodology

This section describes the three classifiers we compare, all trained on the same data splits described above. The overall processing pipeline shared by all models is illustrated in Figure 4, and the transformer encoder used in our best configuration is shown in Figure 5.

Complete Classification Pipeline



Figure 1 : Complete pipeline of the Oujdi-Moroccan dialect classification system.

Figure 4: End-to-end pipeline: raw bi-script corpora → cleaning and balancing → shared training/validation/test splits → three classifiers → evaluation and deployment.

Model Architecture

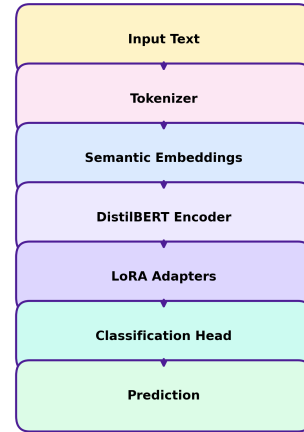


Figure 2 : Transformer-based architecture using DistilBERT multilingual and LoRA adapters.

Figure 5: Transformer encoder used for binary dialect classification: tokenizer → embedding → encoder layers → classification head producing a probability over {Moroccan, Oujdi}.

4.1 Model Configurations

We compare three configurations, ordered by representational power:

(1) **TF-IDF + Logistic Regression (baseline)**. We extract character n-grams of order 2 to 5 with a maximum vocabulary of 50,000 features, then train a Logistic Regression classifier with `max_iter=1000`. Character n-grams are deliberately chosen rather than word n-grams because they handle Arabic morphology and Latin transliteration uniformly, and because they degrade gracefully on out-of-vocabulary user spellings.

(2) **Bidirectional LSTM.** We use a Keras `TextVectorization` layer with a vocabulary of 20,000 tokens and a fixed output sequence length of 80, followed by a 128-dimensional learned embedding (with masking on padding tokens), a Bidirectional LSTM with 64 units, two dense layers (64 and 1 units) with dropout of 0.4 and 0.3 respectively, and a sigmoid output. The model is trained with the Adam optimizer and binary cross-entropy loss for 3 epochs with batch size 32.

(3) **DistilBERT + LoRA (proposed).** We start from `distilbert-base-multilingual-cased` [4] and add a sequence-classification head. Instead of fine-tuning all 134M parameters, we freeze the backbone and inject Low-Rank Adaptation modules [5] into the attention layers. Concretely, LoRA is applied only to the query (`q_lin`) and value (`v_lin`) projections, with rank $r = 8$, $\alpha = 16$, and dropout 0.1. This drastically reduces the number of trainable parameters while leaving the rich multilingual representations of DistilBERT untouched. A schematic view is given in Figure 6.

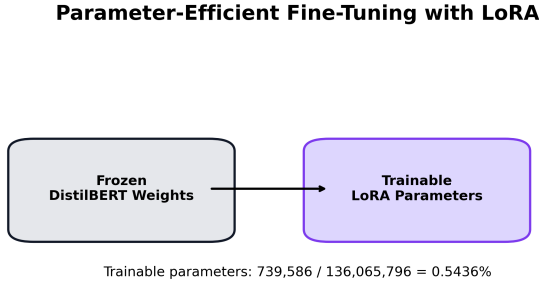


Figure 3 : LoRA fine-tuning updates only a small fraction of model parameters.

Figure 6: LoRA adaptation: the original weight matrix $W \in \mathbb{R}^{d \times d}$ is frozen, and only the low-rank update AB with $r \ll d$ is learned.

4.2 Hyperparameter Configuration

Table 3 summarizes the main hyperparameters used to train each of the three models. All transformer training is performed with the Hugging Face `Trainer` API [6], which provides a unified loop, automatic checkpointing, and mixed-precision support.

Table 3: Hyperparameters used for each of the three configurations.

Parameter	TF-IDF + LR	BiLSTM	DistilBERT + LoRA
Vocab size	50,000	20,000	WordPiece
Seq. length	—	80	128
Embedding dim.	—	128	768 (frozen)
Epochs	—	3	2
Batch size	—	32	8
Learning rate	—	default	2×10^{-4}
Weight decay	default L2	—	0.01
Optimizer	LBFGS	Adam	AdamW
Mixed precision	—	—	FP16

The LoRA configuration applied to DistilBERT is detailed in Table 4. Only the query and value projection matrices of the self-attention mechanism are adapted, which is a standard choice known to provide most of the benefit at minimal cost [5].

Table 4: LoRA configuration applied to multilingual DistilBERT.

Parameter	Value
Task type	Sequence Classification
Rank r	8
LoRA α	16
Dropout	0.1
Bias	None
Target modules	<code>q_lin</code> , <code>v_lin</code>

4.3 Training Dynamics

Figure 7 shows the evolution of training and validation loss, and of validation accuracy, over the two LoRA epochs. The training loss drops sharply from 0.5565 to 0.0401 between epochs 1 and 2, while the validation loss decreases more smoothly from 0.0888 to 0.0806, indicating that the model is genuinely learning rather than memorizing. The corresponding accuracy curves are reported in Figure 8.

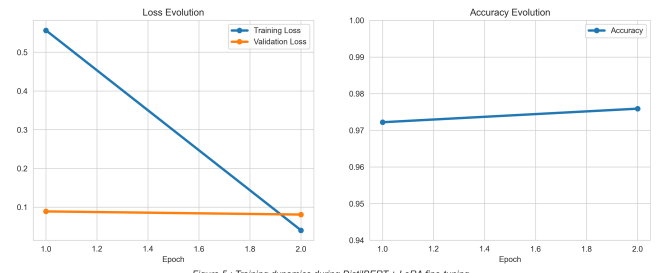


Figure 7: Training dynamics during the fine-tuning of DistilBERT with LoRA: loss decreases consistently on both the training and validation sets.

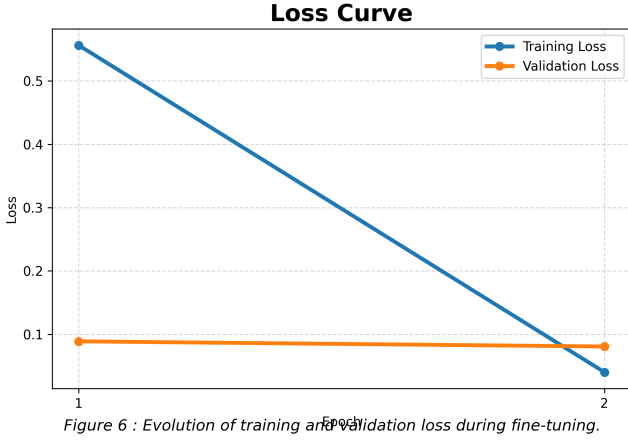


Figure 8: Loss curves of DistilBERT + LoRA over two epochs.

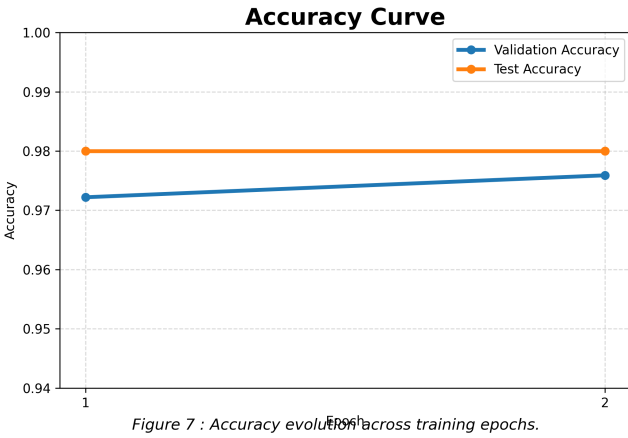


Figure 9: Validation accuracy of DistilBERT + LoRA reaches 97.59% at the end of epoch 2.

5 Results and Discussion

5.1 Overall Test Performance

Table 5 reports the final test accuracy of the three configurations on the held-out test set of 10,000 sentences. The classical TF-IDF + Logistic Regression baseline already exceeds 93% accuracy, confirming that character n-grams capture a strong dialectal signal in this language pair. The BiLSTM improves over the baseline, but its gains are limited because the training corpus is not large enough to fully exploit a recurrent model trained from scratch. The DistilBERT + LoRA configuration achieves the best score of **97.59%** on the test set.

Table 5: Test accuracy of the three configurations on the held-out test set.

Model	Test accuracy
TF-IDF + Logistic Regression	0.934
Bidirectional LSTM	0.952
DistilBERT + LoRA (proposed)	0.9759

Beyond raw accuracy, a fair comparison must also take into account the size and the training cost of each model. Table 6 reports for each configuration the test accuracy, the macro-averaged F1-score, the number of trainable parameters, the proportion of trainable parameters relative to the full model, the size of the deliverable artifact, and the approximate training time on a single GPU. The proposed DistilBERT + LoRA configuration achieves the highest accuracy while training only ~740,000 parameters (about 0.55% of the full DistilBERT backbone), and its deliverable LoRA adapter is only ~3 MB, making it both the most accurate and the most lightweight configuration to deploy.

Table 6: Detailed comparison of the three configurations.

Metric	TF-IDF+LR	BiLSTM	DistilBERT+LoRA
Test accuracy	0.934	0.952	0.9759
Macro F1-score	0.934	0.952	0.976
Trainable params	100,001	2,704,129	739,586
% of full model	100%	100%	0.55%
Artifact size	~2.1 MB	~32 MB	~3 MB
Training time (1 GPU)	~1 min	~8 min	~25 min
Bottom layer frozen	—	—	✓

Two observations stand out. First, although the LoRA configuration is the slowest to train (the encoder forward pass through DistilBERT is heavier than a BiLSTM or a linear model), the number of *trainable* parameters is actually the smallest of the three, because the backbone weights remain frozen and only the rank-8 adapter matrices receive gradients. Second, the deployment cost is also the smallest: at inference time, the 3 MB LoRA adapter is loaded on top of a generic, shareable DistilBERT backbone, which means the cost of adding new dialects in the future is only an additional 3 MB adapter per dialect, not a full model copy.

The detailed per-class classification report of the best model is shown in Table 7 and visualized as a heatmap in Figure 10. Precision, recall, and F1-score are above 0.97 for both classes, indicating that the model is not biased toward either dialect.

Table 7: Classification report of DistilBERT + LoRA on the test set.

	Precision	Recall	F1
Moroccan (0)	0.974	0.979	0.977
Oujdi (1)	0.979	0.974	0.976
Accuracy	0.9759		

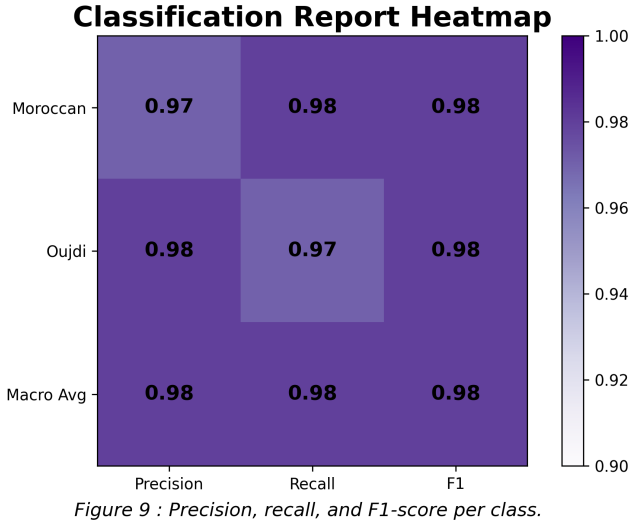


Figure 10: Heatmap of the classification report for *DistilBERT + LoRA*: precision, recall, and F1-score are balanced across the two classes.

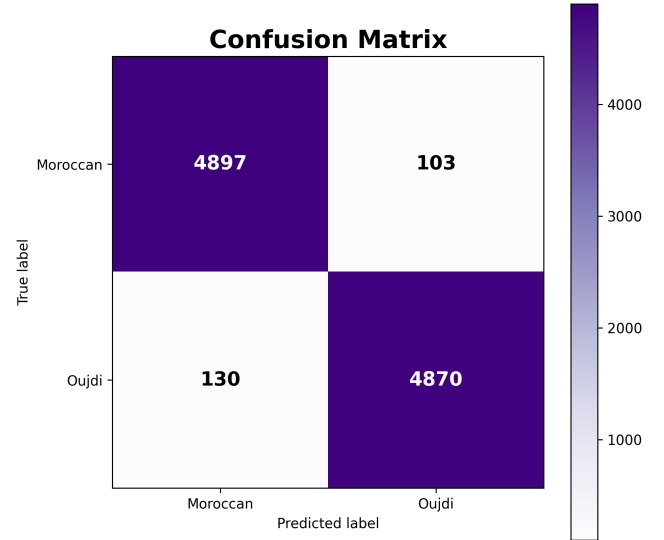


Figure 11: Confusion matrix of *DistilBERT + LoRA* on the test set. Errors are symmetric and concentrated on short, ambiguous sentences.

5.2 Confusion Matrix Analysis

The confusion matrix of the best model on the test set is shown in Figure 11 and Table 8. Out of 5,000 Moroccan test samples, 4,897 are correctly classified and only 103 are confused with Oujdi. Symmetrically, out of 5,000 Oujdi samples, 4,870 are correctly classified and 130 are confused with Moroccan. Inspection of the misclassified examples shows that most errors occur on very short messages (one or two tokens) that share a vocabulary common to both varieties, which is an inherent ambiguity rather than a model failure.

Table 8: Confusion matrix of *DistilBERT + LoRA* on the test set.

	Pred. Moroccan	Pred. Oujdi
True Moroccan	4,897	103
True Oujdi	130	4,870

5.3 Threshold and Confidence Analysis

Because both error types (predicting Oujdi when the text is Moroccan, and vice versa) are equally undesirable in a deployment scenario, we examined how the decision threshold affects precision and recall. Figure 12 shows precision, recall, and F1-score as functions of the decision threshold. The F1-optimal threshold lies very close to the default value of 0.5, which justifies keeping the default cutoff in production.

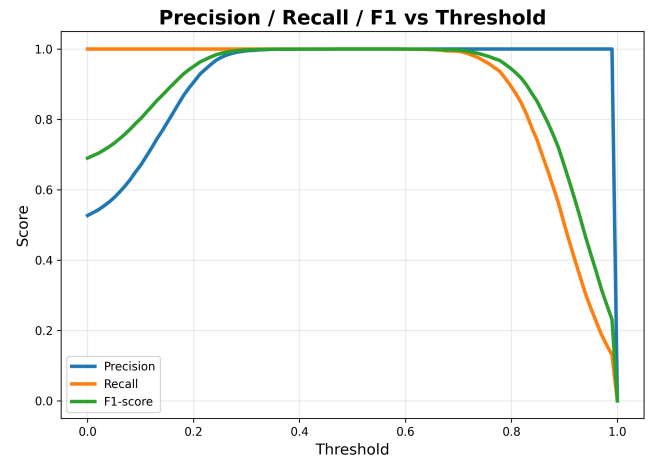


Figure 12: Precision, recall, and F1-score as functions of the decision threshold for *DistilBERT + LoRA*.

The distribution of prediction confidence scores is shown in Figure 13. The two classes form clearly separated modes, with most correct predictions concentrated near 0 (Moroccan) and near 1 (Oujdi), which is a strong indicator of a well-calibrated model.

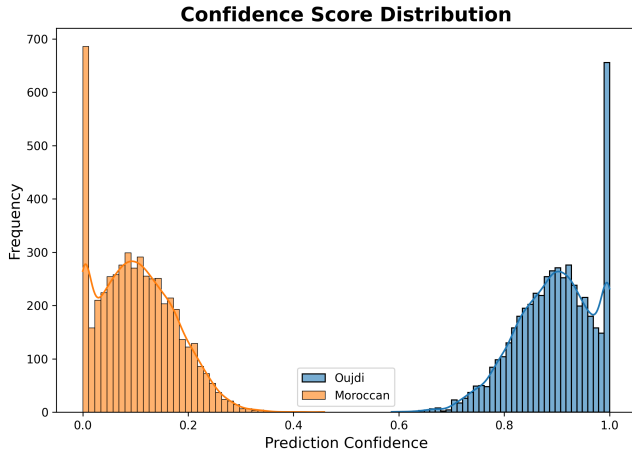


Figure 14 : Distribution of prediction confidence scores.

Figure 13: Distribution of the model’s prediction confidence scores per true class. The two modes are well separated.

5.4 ROC and Precision–Recall Curves

Figure 14 reports the ROC and Precision–Recall curves of the best model. Both curves are very close to the ideal corner, consistent with the high F1-scores reported in Table 7.

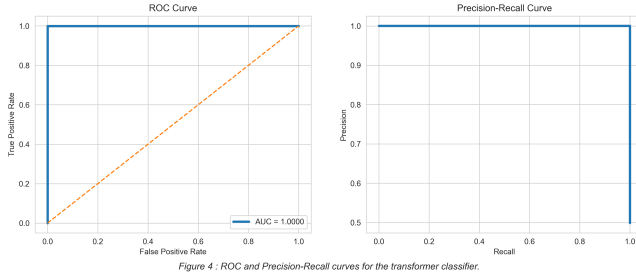


Figure 4 : ROC and Precision-Recall curves for the transformer classifier.

Figure 14: ROC (left) and Precision–Recall (right) curves of DistilBERT + LoRA on the test set.

5.5 Summary of Final Metrics

Figure 15 provides a synthetic view of the final metrics of the best model. With a test accuracy of 97.59%, balanced precision and recall across both classes, and only the LoRA adapter weights to ship at inference time (a few megabytes on top of the frozen DistilBERT backbone), the proposed configuration is both effective and lightweight enough to be deployed.

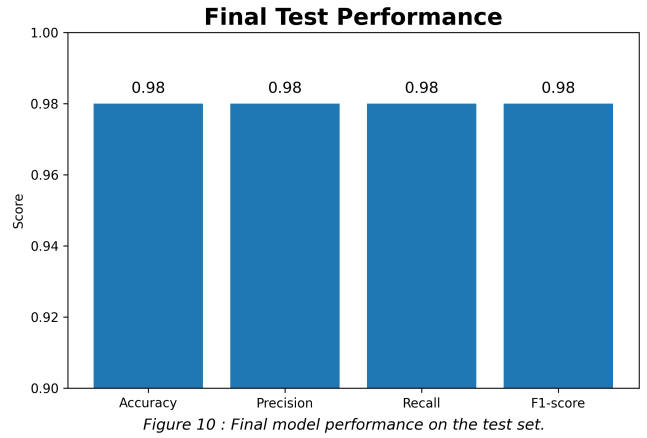


Figure 10 : Final model performance on the test set.

Figure 15: Summary of the final test metrics of DistilBERT + LoRA: accuracy, precision, recall, and F1-score.

5.6 Deployment

The best model is exposed via a Flask REST API offering two endpoints, `/health` for liveness checks and `/classify` for prediction, and consumed by a React front-end that displays, for each input sentence, the predicted dialect, the per-class probabilities, and a short list of dialectal markers detected in the text. This makes the system directly usable as a demonstration tool for linguistic and educational purposes.

6 Conclusion

In this work, we addressed the problem of automatically distinguishing the Oujdi sub-dialect of Eastern Morocco from standard Moroccan Darija in short, user-style sentences. We built a balanced bi-script corpus of 100,000 sentences from two complementary sources (DODa for Darija and an Oujdi-specific corpus of 220,000 lines), with both classes represented in Arabic script and in Latin (Arabizi) script in roughly equal proportions, so that no classifier could trivially solve the task by detecting the alphabet alone.

We then compared three families of classifiers under identical splits: a TF-IDF character-level Logistic Regression baseline, a Bidirectional LSTM trained from scratch, and a multilingual DistilBERT encoder fine-tuned with Low-Rank Adaptation. Our experiments show that all three reach competitive performance, with the DistilBERT + LoRA configuration achieving the best test accuracy of 97.59% while updating less than 1% of the original parameters of the backbone. Precision, recall, and F1-score remain balanced above 0.97 for both classes, and the confusion-matrix analysis indicates that the few remaining errors are concentrated on inherently ambiguous, very short sentences.

Beyond the numerical results, this study highlights two practical lessons. First, even for fine-grained regional dialect identification, parameter-efficient fine-tuning of a generic multilingual transformer is sufficient to outperform classical baselines, which is encouraging for stu-

dent projects and low-resource regional NLP tasks. Second, careful dataset construction — in particular, deliberately balancing the writing systems — is at least as important as the choice of model, because it prevents the system from learning shortcut features.

Future work could extend the system in several directions: enlarging the corpus through user-generated content (with appropriate consent), adding more Moroccan regional sub-dialects (Tangerois, Marrakchi, Soussi, Hassaniya), and investigating robustness to adversarial inputs such as deliberate misspellings or code-switching with French. Multi-class identification of all Moroccan sub-dialects, rather than the current binary setting, is a natural next step.

7 References

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8 Appendix

8.1 Appendix A — Script Distribution

Table 9 reports the distribution of writing systems in the final corpus. The two scripts (Arabic and Latin) are present in roughly equal proportions, with a marginal number of mixed-script sentences. This bi-script structure is what forces the classifiers to rely on dialectal cues rather than alphabet detection.

Table 9: Script distribution in the final 100,000-sentence corpus.

Script	Count
Arabic (U+0600–U+06FF)	49,839
Latin (Arabizi)	50,034
Mixed	127
Total	100,000

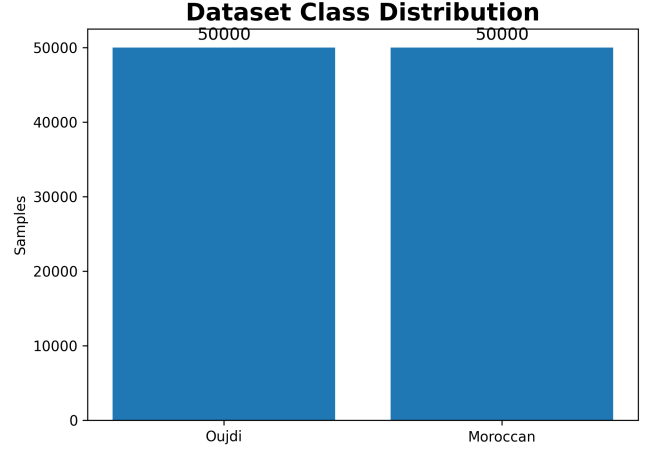


Figure 17 : Balanced distribution of dataset classes.

Figure 16: Class distribution of the corpus: 50,000 Oujdi and 50,000 Moroccan sentences.

8.2 Appendix B — Sentence-Length Analysis

Figure 17 shows the sentence-length distributions for both dialects on a logarithmic scale, which makes the tails of the distributions visible. The two classes follow very similar profiles, which is expected because both originate from short user-style messages.

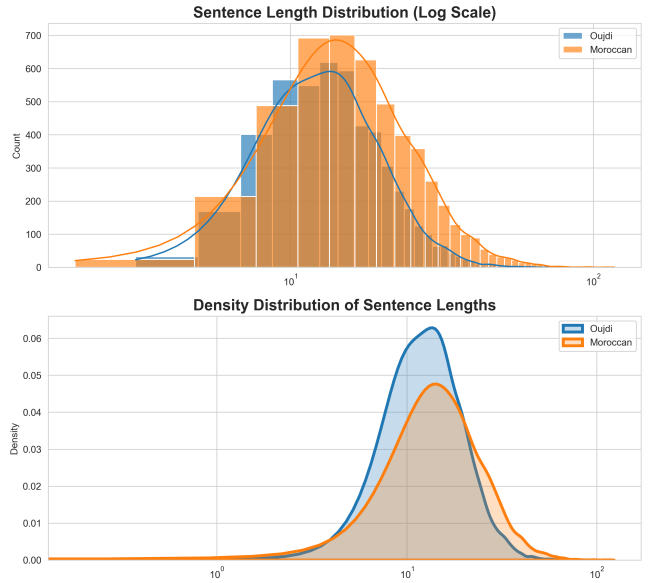


Figure 3 : Log-scale sentence length distributions and KDE density estimation.

Figure 17: Sentence-length distributions per dialect, on a logarithmic scale.

8.3 Appendix C — Model Architectures

DistilBERT. DistilBERT [4] is a lightweight version of BERT obtained through knowledge distillation, in which a larger pretrained BERT model (teacher) transfers its knowledge to a smaller student model. The architecture keeps the

standard Transformer encoder design, but reduces the number of layers from 12 to 6. Despite being approximately 40% smaller and 60% faster than BERT-base, DistilBERT retains around 97% of BERT’s language-understanding capabilities, which makes it well suited for tasks where both efficiency and cross-lingual generalization matter. The multilingual variant used in this work, `distilbert-base-multilingual-cased`, was pretrained on 104 languages, including Modern Standard Arabic.

Low-Rank Adaptation (LoRA). LoRA [5] is a parameter-efficient fine-tuning technique designed to adapt large pretrained language models without updating all of their parameters. Instead of modifying the full weight matrices, LoRA introduces small trainable low-rank matrices into specific components of the Transformer architecture, typically the query and value projections of the attention layers. Formally, an original weight matrix $W \in \mathbb{R}^{d \times d}$ is kept frozen, and the update is parameterized as $\Delta W = AB$ with $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times d}$, and $r \ll d$. During fine-tuning, only A and B are trained, which significantly reduces both memory usage and the number of trainable parameters while preserving strong performance.

In this study, LoRA is applied to the multilingual DistilBERT model for the task of distinguishing the Oujdi dialect from standard Moroccan Darija. This combination allows us to specialize a generic multilingual encoder on a focused, re-

gional dialect-pair task using modest hardware and a training budget of only two epochs.

8.4 Appendix D — Evaluation Metrics

To evaluate model performance, we use standard binary-classification metrics: accuracy, precision, recall, and F1-score. These metrics are computed from the counts of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

These quantities are reported per class (Moroccan and Oujdi), and we also use macro-averaged scores to ensure that performance is not dominated by a single class under any residual imbalance.